13 COMPARISON: MIDTERM FRAME (A) vs POLYFRAME NET (19), SAME LOADS

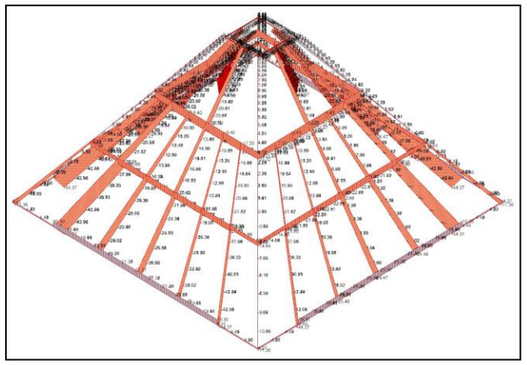
Same casing weight (printed shell as weight only: 501.6 kN midterm, 500 kN here) and the same typhoon (qp 2.6 kN/m2, cpe +0.7 / -0.6 / -0.7 on the four roof faces). LONG = 1.0 G, SHORT = 1.0 G + 1.0 W.
Left of each pair: the midterm frame (columns, hips, rings; from the midterm page 05). Right: the form-found PolyFrame net 19 (568 cast bars 150 x 250, 3 m crown ring, 24 supports).

AXIAL FORCE Nx

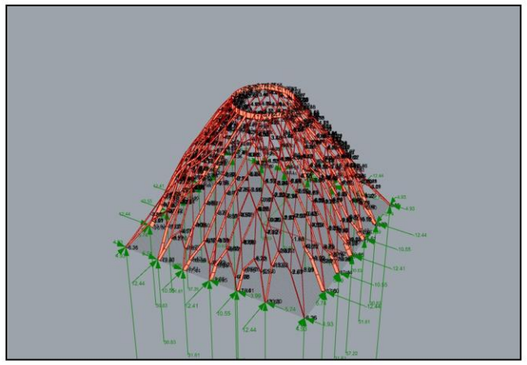
A MIDTERM FRAME

19 POLYFRAME NET

LONG

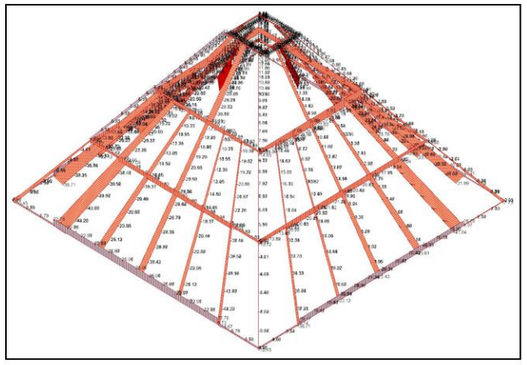


columns -44.4 kN at the feet; base ring +19 to +21 kN (tension)

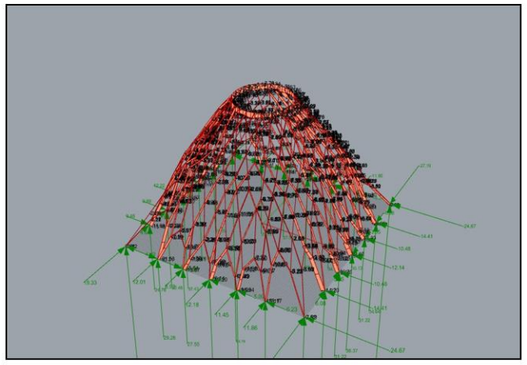


-27 kN at most (crown ring); no real tension (+0.4 kN)

SHORT



columns down to -43 kN; base ring up to +26 kN



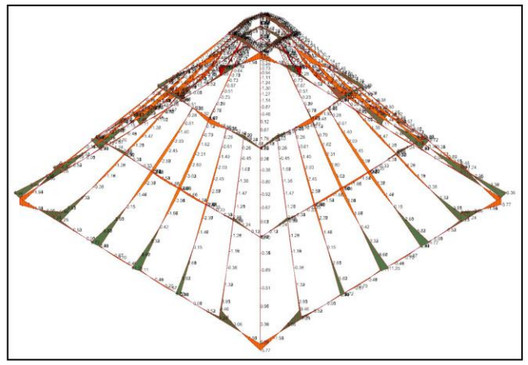
-31 kN at most; +4.8 kN in a few upper bars

BENDING My

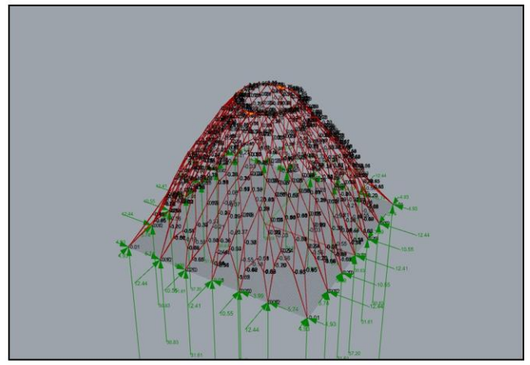
A MIDTERM FRAME

19 POLYFRAME NET

LONG

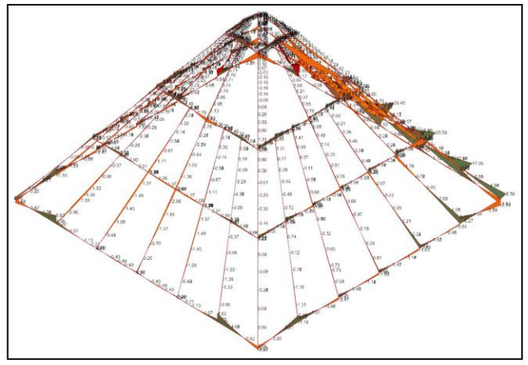


11.25 kNm at every column foot

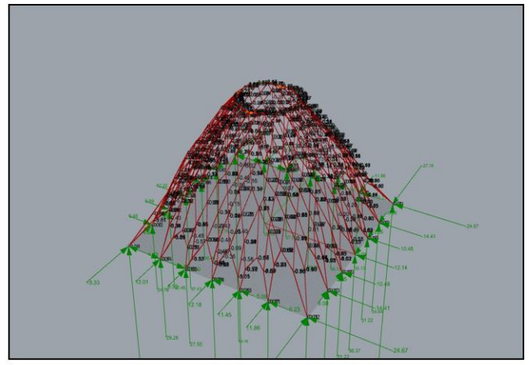


under 1.2 kNm in the net; 4.1 kNm in the crown ring

SHORT



26.45 kNm at the windward column feet (sizes the columns)



under 1.4 kNm in the net; 4.3 kNm in the crown ring

SIDE BY SIDE (LONG / SHORT)

	A MIDTERM FRAME	19 POLYFRAME NET
Cast concrete above the supports	9.0 m3 (ribs + rings)	21.9 m3 (568 bars 150 x 250)
Roof weight carried (casing + cast)	717.6 kN	1024.7 kN
Max compression N	-44.4 / -44.8 kN (columns)	-27.3 / -30.8 kN (crown ring)
Max tension N	+21.5 / +26.0 kN (base ring)	+0.4 / +4.8 kN (a few upper bars)
Max bending My	11.2 / 26.4 kNm (column feet)	1.2 / 1.3 kNm net; 4.1 / 4.3 kNm crown ring
Max sideways bending Mz	14.1 / 14.7 kNm (middle rings)	1.0 / 1.5 kNm net; 1.4 / 2.0 kNm crown ring
Total wind push, SHORT	116.5 kN, 20 footings	112.7 kN, 24 supports
Max deflection	-	0.3 / 0.5 mm

WHAT IT SHOWS

The net carries the roof by compression: its bars bend 10 to 20 times less than the frame columns.

It needs no tension tie inside the roof (A: base ring +26 kN).

It uses 2.4 x more cast concrete, and the crown ring is the one member that bends (4 kNm).

Both push out on their supports under the typhoon: ~113 kN in total.

Not modelled for 19: base ring, buttresses, footings.